LAB EXPERIMENT RECORD

Set 23: Employee Performance analysis

Department	CSE-DS
Course Code / Course Name	DSA0601 / Data Handling and Visualization
Experiment	Data Visualization using R
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Day	Saturday

1. Bar Chart

Aim

To create a bar chart in R to visualize the distribution of employees across different departments and label the chart elements.

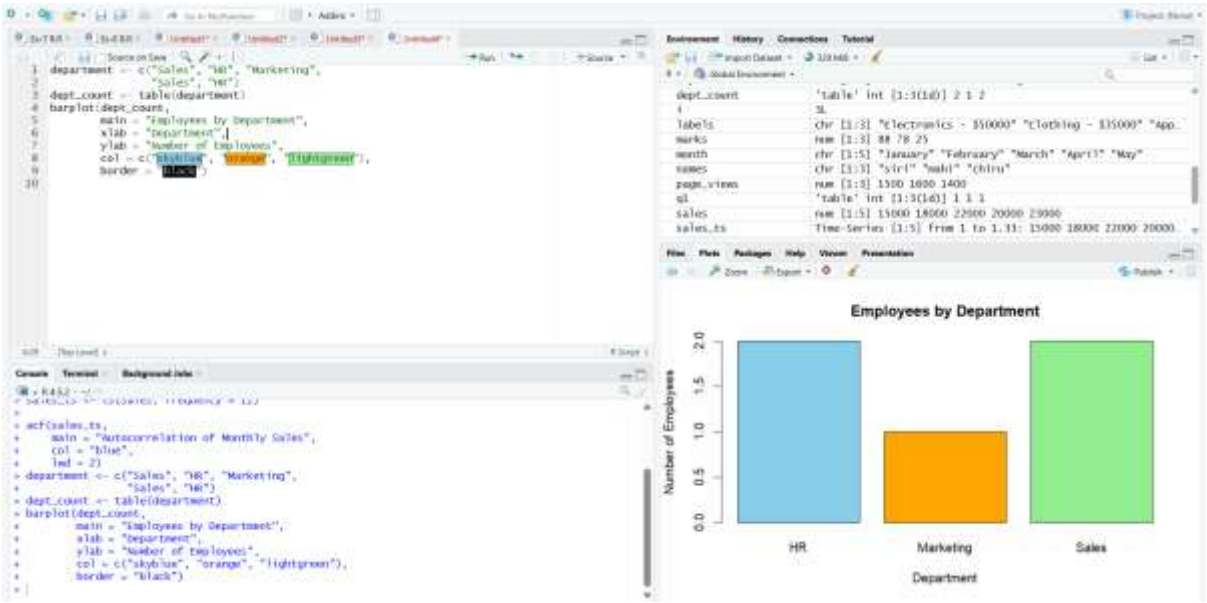
R Code

R

```
department <- c("Sales", "HR", "Marketing",
                "Sales", "HR")
dept_count <- table(department)
barplot(dept_count,
    main = "Employees by Department",
    xlab = "Department",
    ylab = "Number of Employees",
    col = c("skyblue", "orange", "lightgreen"),
```

border = "black")

Output



A bar chart showing the number of employees in each department.

Observation

Sales and HR each have 2 employees, while Marketing has 1 employee. Sales and HR have the highest employee count.

2. Line Chart

Aim

To create a line chart in R to visualize employee performance scores and label the axes and title the chart.

R Code

R

employee_id <- c(1, 2, 3, 4, 5)

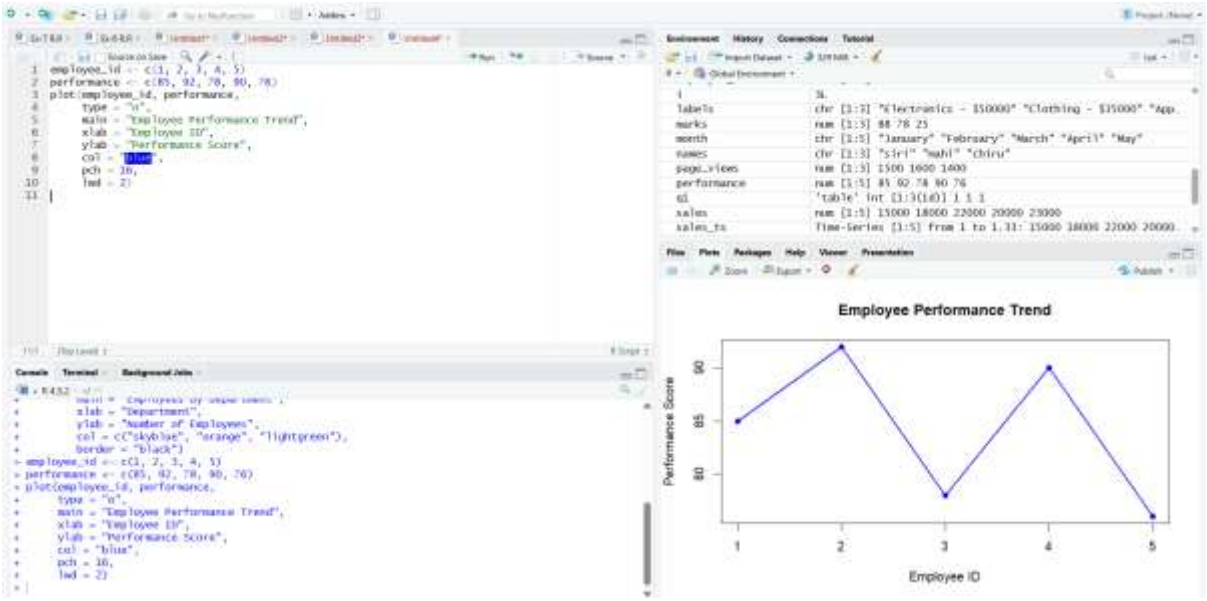
performance <- c(85, 92, 78, 90, 76)

plot(employee_id, performance,

type = "o",

```
    main = "Employee Performance Trend",
    xlab = "Employee ID",
    ylab = "Performance Score",
    col = "blue",
    pch = 16,
    lwd = 2)
```

Output



A line chart showing the performance scores of five employees.

Observation

Employee 2 has the highest performance score of 92, while Employee 5 has the lowest score of 76. The scores vary across employees.

3. Tableau Dashboard

Aim

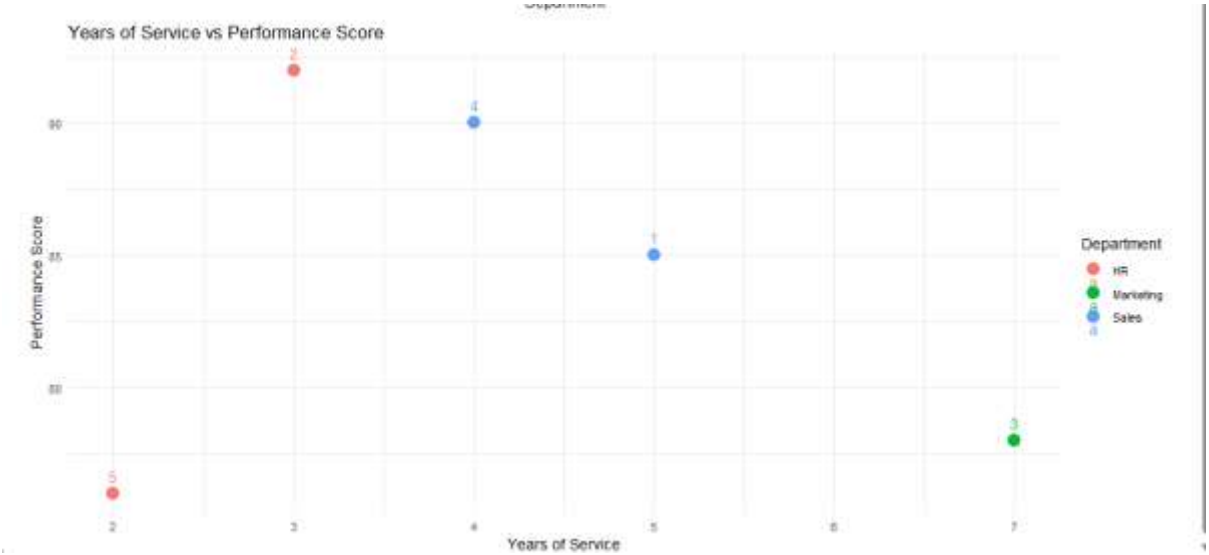
To develop a dashboard combining a bar chart showing department distribution and a scatter plot displaying the relationship between years of service and performance scores.

Tableau Procedure

1. Open Tableau and import the Employee Performance dataset.
2. Create a bar chart using Department and Employee ID.
3. Set Employee ID to Count to display the number of employees in each department.
4. Add a suitable chart title and axis labels.
5. Create a scatter plot using Years of Service on the X-axis and Performance Score on the Y-axis.
6. Use Employee ID to distinguish individual employees.
7. Assign different colors to departments.
8. Select Dashboard and create a new dashboard.
9. Drag the department bar chart into the dashboard.
10. Drag the scatter plot into the dashboard.
11. Arrange both charts neatly and add suitable titles.
12. Save the dashboard.

Output





A dashboard displaying employee distribution across departments and the relationship between years of service and performance scores.

Observation

The dashboard helps compare department sizes and examine how employee performance scores vary with years of service.

4. Employee Performance Table

Aim

To develop a table in R displaying employee IDs, departments, years of service, and performance scores.

R Code

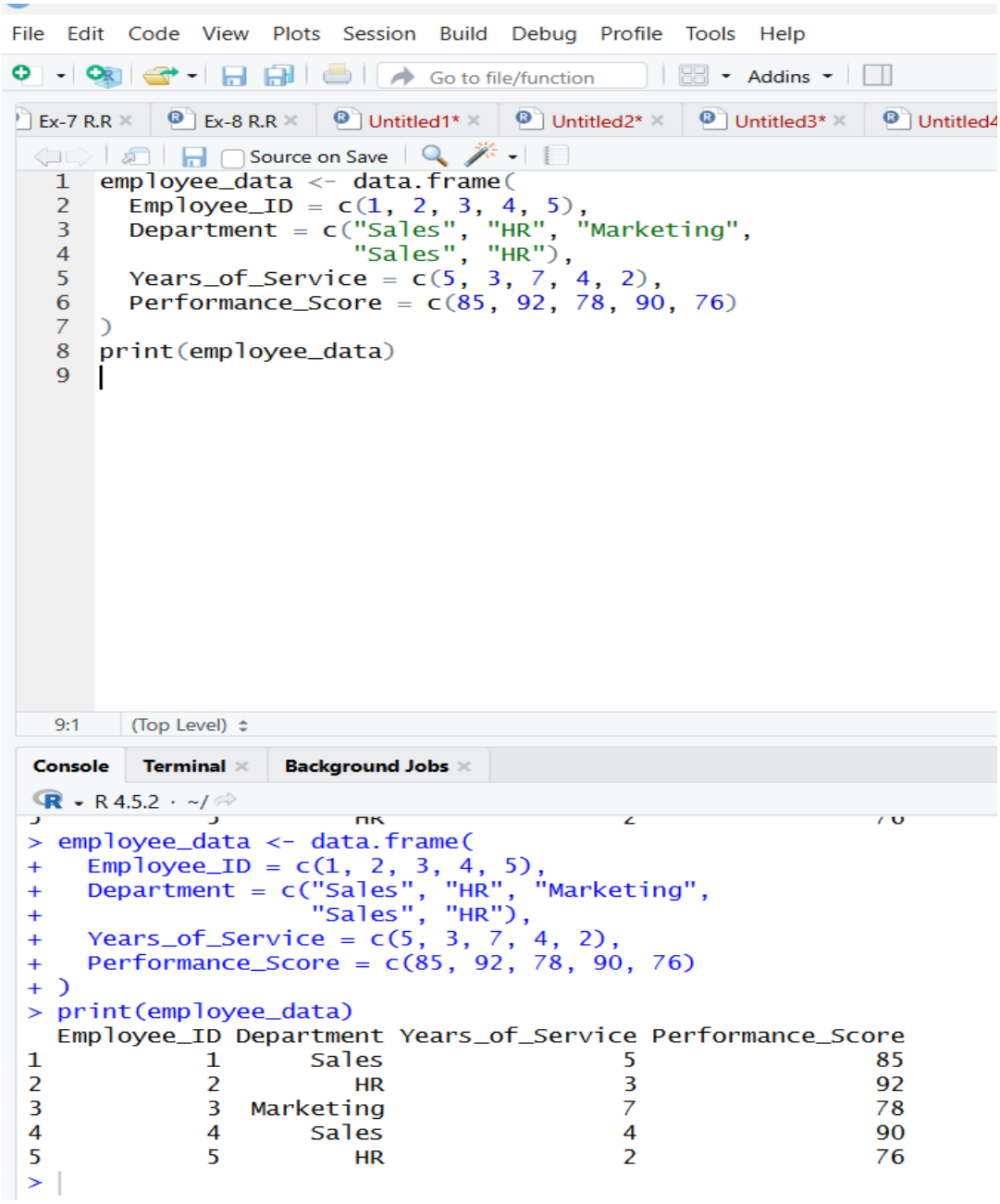
R

```
employee_data <- data.frame(
  Employee_ID = c(1, 2, 3, 4, 5),
  Department = c("Sales", "HR", "Marketing",
                 "Sales", "HR"),
  Years_of_Service = c(5, 3, 7, 4, 2),
  Performance_Score = c(85, 92, 78, 90, 76)
```

)

print(employee_data)

Output



A table displaying the complete performance details of all five employees.

Observation

The table clearly presents employee details and performance scores. Employee 2 has the highest score of 92, and Employee 5 has the lowest score of 76.

Result

The employee performance dataset was successfully analyzed using a bar chart, line chart, and data table in R. A Tableau dashboard was also outlined to compare department distribution and employee performance against years of service.